

# DAYANANDA SAGAR UNIVERSITY, BENGALURU

School of Engineering • End Semester Examination

Course: Analog and Digital Electronics | Course Code: 21CS33 | Credits: 3

Exam Session: End Semester Exam 2023

Max Marks: 80 | Duration: 3 Hours

## INSTRUCTIONS TO CANDIDATES:

1. Answer FIVE full questions, choosing ONE full question from each module.
2. Draw neat diagrams wherever necessary. Missing data may be suitably assumed.

### MODULE 1

**Q1.** Simplify the Boolean function  $F(A, B, C, D) = \sum m(0, 2, 5, 7, 8, 10, 14, 15) + \sum d(3, 11)$  using a 4-variable Karnaugh Map and realize the simplified expression using basic logic gates. [8 Marks]

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**Q2.** Design a 3-bit Synchronous Up-Counter using JK Flip-Flops. Draw the state diagram, excitation table, K-maps for J & K inputs, and final logic schematic. [8 Marks]